

OM NATHWANI

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Education

University of Waterloo & Wilfrid Laurier University

BCS & BBA, Computer Science and Business Administration (Co-op)

Waterloo, Ontario

2024 - 2029

- **GPA:** 3.7/4.0
- **Extracurricular:** Dual-Varsity Athlete in Cross Country and Track ↗, President of Waterloo Hindu Yuva ↗
- **Awards:** 2025 USports Academic All-Canadian ↗, BDO Future Leaders Challenge Finalist ↗

Technical Skills

- **Languages:** Python, C++, C, Java, JavaScript, TypeScript, SQL, HTML/CSS
- **Libraries/Frameworks:** Flask, FastAPI, GraphQL, SQLAlchemy, Apache Airflow, PyTorch, TensorFlow, Pandas, NumPy, LangChain, ChromaDB, scikit-learn, Express.js, Node.js, Next.js, React.js, Tailwind CSS
- **Tools:** PostgreSQL, Linux, Docker, Git, GitHub Actions, GCP, AWS, Supabase, LangSmith, Kyriba, Shopify GraphQL API

Experience

Software Developer

Dawson Partners | *Python, FastAPI, SQLAlchemy, GCP, Apache Airflow, LangSmith, Pandas, NumPy, Kyriba, Git*

Jan 2026 – Apr 2026

Toronto, Ontario

- Built the **Kyriba** submission workflow for Dawson's **internal expense management platform**, routing invoice and intercompany rebillings by payment type and mapping account data into records used to process **200+** weekly corporate payments
- Implemented **idempotent Kyriba** payment submissions, storing a unique reference in both Kyriba and the database to prevent duplicate transactions, keeping linked invoice and intercompany rebilling records synchronized through **status polling**
- Rearchitected Dawson's **K-1 tax document pipeline** using page-level **SQLAlchemy** models and **Apache Airflow** workflows for **270** fields, preserving full edit history and integrating documents into a shared architecture reusable across additional tax forms
- Built validation and structured error-handling workflows for Dawson's **financial forecasting models**, increasing the pass rate from **27%** to **98%** across our internal test suite, and returning actionable errors that prevented agents from repeating model calls
- Developed a **LangSmith** trace-analysis workflow to map **380+** inputs, outputs, and variables across Dawson's **agent-driven financial forecasting system**, surfacing data-quality and logic gaps that limited scenario construction and result interpretation

Software Engineer

Covalense Digital | *Python, FastAPI, CLIP, pdfplumber, Pytesseract, PostgreSQL, Psycopg2, ChromaDB, Ollama, OCR, Git*

May 2025 – August 2025

Mississauga, Ontario

- Built, shipped, and presented a secure enterprise document assistant to **senior executives**, implementing an **MCP** retrieval layer with document-ingestion and **RBAC-filtered ChromaDB** semantic-search tools across **500+** documents and **1,000+** images
- Extended the product's document workflow into a **multimodal retrieval pipeline** that extracted PDF images, generated **CLIP** embeddings, and linked vector metadata to source assets so the agent could return and analyze the exact image matched to a query
- Evaluated **50+** document-processing and vector-search libraries against licensing, performance, and integration requirements, selecting appropriately licensed tools to reduce third-party risk and support **commercial enterprise deployment**

Backend Developer

Lazaridis Students' Society | *Express.js, Next.js, React.js, Shopify, Google OAuth, GraphQL*

May 2026 – Ongoing

Waterloo, Ontario

- Built a modular internal platform for **23 student organizations** using **Next.js**, **Express**, and **TypeScript**, creating a **scalable base** for applications that reduce manual support and centralize internal operations across the Lazaridis Students' Society
- Developed an analytics application of **Shopify** product data, using **GraphQL** pipelines, giving clubs self-serve access to sales, inventory, refunds, and revenue reporting, eliminating **90+** annual manual data requests
- Implemented **multi-level RBAC** with server-side sessions, account and application-level permissions, and club-scoped data filtering, preventing cross-club and cross-role access to restricted tools and commerce data
- Built workflows to join product, purchaser, and order-level data into **standardized financial records**, enabling student organizations to reconcile sales activity, and reuse data for downstream financial reporting

Projects

Bollyguess ↗ | *Next.js, Node.js, Tailwind CSS, React, Python, PostgreSQL, Supabase, Docker, GitHub Actions, Vercel* **July 2025**

- Launched and operated a daily Bollywood song-guessing platform with **Next.js** and **PostgreSQL**, reaching **1,000+** users
- Designed a normalized **PostgreSQL** schema and ingestion workflows for **2,400+** songs and **7,500+** artists, supporting **many-to-many** relationships and **scalable** content expansion
- Automated daily game generation using **Python** selection logic, relational **PostgreSQL** queries across **9,000+** records, and scheduled **GitHub Actions**, enabling a new game to be published each day without recurring manual updates

Audio Segment Classifier ↗ | *Python, PyTorch, TensorFlow, Librosa, FFmpeg, Pandas, scikit-learn*

July 2025

- Built a batch preprocessing and feature-extraction pipeline for **2,400+** audio segments using **FFmpeg**, **Librosa**, and **Pandas**, transforming unstructured audio into a structured dataset of **27** temporal and spectral features for automated predictive scoring
- Trained and tuned a **PyTorch feedforward neural network** with **four hidden layers** to classify audio segments as suitable or unsuitable for gameplay, achieving **93%** validation accuracy and approximately **0.82 macro F1**
- Integrated the classifier into **BollyGuess's daily game-generation workflow**, running inference across 10-second song segments and using the resulting suitability scores to rank candidates and select the highest-quality clip for each game